

Together, neuron development, neuron death, and synapse elimination establish the basic network of cells and connections within the nervous system required throughout life.

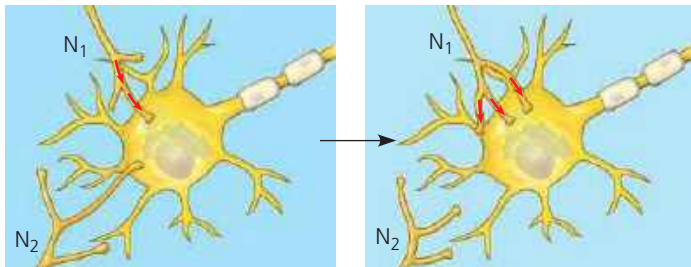
Neuronal Plasticity

Although the overall organization of the CNS is established during embryonic development, the connections between neurons can be modified. This capacity for the nervous system to be remodeled, especially in response to its own activity, is called **neuronal plasticity**.

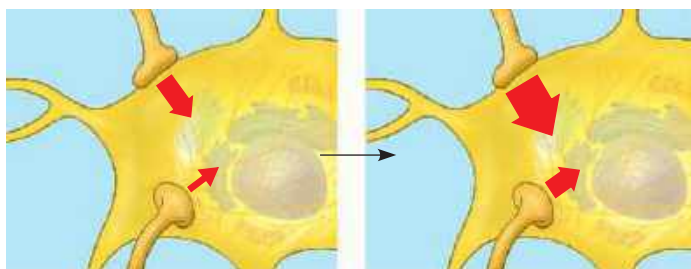
Much of the reshaping of the nervous system occurs at synapses. Synapses belonging to circuits that link information in useful ways are maintained, whereas those that convey bits of information lacking any context may be lost. Specifically, when the activity of a synapse coincides with that of other synapses, changes may occur that reinforce that synaptic connection. Conversely, when the activity of a synapse fails to coincide with that of other synapses, the synaptic connection sometimes becomes weaker.

Figure 49.21a illustrates how activity-dependent events can trigger the gain or loss of a synapse. If you think of signals in the nervous system as traffic on a highway, such changes are comparable to adding or removing an entrance ramp. The net effect is to increase signaling between particular pairs of neurons and decrease signaling between other pairs. Signaling at a synapse can also be strengthened or weakened, as

▼ **Figure 49.21 Neuronal plasticity.** Synaptic connections can change over time, strengthening or weakening in response to the level of activity at the synapse.



(a) High-level activity at the synapse of neuron N_1 with the postsynaptic neuron leads to recruitment of additional axon terminals from that neuron. Lack of activity at the synapse with neuron N_2 leads to loss of functional connections with that neuron.



(b) If two synapses on a postsynaptic cell are often active at the same time, the strength of both responses may increase.

shown in **Figure 49.21b**. In our traffic analogy, this would be equivalent to widening or narrowing an entrance ramp.

A defect in neuronal plasticity may underlie *autism spectrum disorder*, which results in impaired communication and social interaction, as well as stereotyped and repetitive behaviors beginning in early childhood. There is now growing evidence that autism spectrum disorder involves a disruption of activity-dependent remodeling at synapses. At the same time, extensive research has ruled out any link to vaccine preservatives, once proposed as a potential risk factor based on fraudulent data.

Although the underlying causes of autism are unknown, there is a strong genetic contribution to this and related disorders. Further understanding of the autism-associated disruption in synaptic plasticity may help efforts to better understand and treat this disorder.

Memory and Learning

Neuronal plasticity is essential to the formation of memories. We are constantly checking what is happening against what just happened. We hold information for a time in **short-term memory** and then release it if it becomes irrelevant. If we wish to retain knowledge of a name, phone number, or other fact, the mechanisms of **long-term memory** are activated. If we later need to recall the name or number, we fetch it from long-term memory and return it to short-term memory.

Short-term and long-term memory both involve the storage of information in the cerebral cortex. In short-term memory, this information is accessed via temporary links formed in the hippocampus. When memories are made long-term, the links in the hippocampus are replaced by connections within the cerebral cortex itself. As discussed earlier, some of this consolidation of memory is thought to occur during sleep. Furthermore, the reactivation of the hippocampus that is required for memory consolidation likely forms the basis for at least some of our dreams.

According to our current understanding of memory, the hippocampus is essential for acquiring new long-term memories but not for maintaining them. This hypothesis readily explains the symptoms of some individuals who suffer damage to the hippocampus: They cannot form any new lasting memories but can freely recall events from before their injury. In effect, their lack of normal hippocampal function traps them in their past. Hippocampal damage and memory loss are common in the early stages of Alzheimer's disease (see Concept 49.5).

What evolutionary advantage might be offered by organizing short-term and long-term memories differently? One hypothesis is that the delay in forming connections in the cerebral cortex allows long-term memories to be integrated gradually into the existing store of knowledge and experience, providing a basis for more meaningful associations.